# SQL Practice Questions

1. SELECT * FROM

- Fetch all details of employees who belong to the 'Sales' department.

```
SELECT * FROM employees WHERE department = 'Sales';
```

2. SELECT DISTINCT

- List all unique payment modes used in transactions.

```
SELECT DISTINCT payment_mode FROM transactions;
```

3. COUNT

- Count the number of employees in each department.

```
SELECT department, COUNT(*) AS total_employees FROM employees GROUP BY department;
```

4. Alias (AS)

- Display employee names and their salary incremented by 10% as "New Salary".

```
SELECT emp_name, salary * 1.10 AS "New Salary" FROM employees;
```

5. WHERE

- Get details of students who scored above 80 and are in grade 'A'.

```
SELECT * FROM students WHERE score > 80 AND grade = 'A';
```

6. LIMIT and ORDER BY

- Retrieve the names of the top 3 highest scoring students.

```
SELECT student_name, score FROM students ORDER BY score DESC LIMIT 3;
```

7. Operators and Filtering

- Get the employees who are either in 'HR' or 'IT' department and earn more than 70,000.

SELECT * FROM employees WHERE department IN ('HR', 'IT') AND salary > 70000;

8. BETWEEN, IN, LIKE, and ILIKE

- Fetch products with prices between 100 and 500.

SELECT * FROM products WHERE price BETWEEN 100 AND 500;

9. CASE, IF, ELSE

- Show the product name and its availability status as 'In Stock' if quantity > 0, else 'Out of Stock'.

SELECT product_name, CASE WHEN quantity > 0 THEN 'In Stock' ELSE 'Out of Stock' END AS availability FROM inventory;

10. COALESCE and NULLIF

- Display customer names and delivery dates, but if the delivery date is NULL, show 'Not Delivered'.

SELECT customer_name, COALESCE(delivery_date, 'Not Delivered') AS delivery_status FROM orders_v2;

Advanced Challenges:

1. Combining Concepts:

- Retrieve all products whose names contain 'a' (case insensitive) and are priced between 50 and 200, ordered by price in ascending order.

SELECT * FROM products WHERE product_name ILIKE '%a%' AND price BETWEEN 50 AND 200 ORDER BY price ASC;

2. Using CASE with Aggregate:

- Count the number of students in each grade and label it as 'High Count' if more than 1 student, otherwise 'Low Count'.

SELECT grade, COUNT(*) AS student_count, CASE WHEN COUNT(*) > 1 THEN 'High Count' ELSE 'Low Count' END AS count_category FROM students GROUP BY grade;

3. Using COALESCE and NULLIF Together:

- Display employee names and their bonus (salary * 0.1). If the salary is NULL, show the bonus as 0. If the bonus is 0, show it as NULL.

SELECT emp_name, NULLIF(COALESCE(salary * 0.1, 0), 0) AS bonus FROM employees;